# SHALEM NALLI

**Frontend Engineer / Full-Stack Engineer**

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## PROFESSIONAL SUMMARY

Frontend engineer with production experience building React and TypeScript applications, including Next.js apps with server-side rendering. I build reusable component libraries and design systems, manage state with Redux and Zustand, and integrate REST APIs into production interfaces. I write unit and end-to-end tests with Jest, React Testing Library, and Playwright, and pay attention to performance and responsive design. I work directly with product and design teams in Agile/Scrum settings and take features from requirements through to production. Alongside frontend work, I build backend services in Python/FastAPI and Node.js, deploy on Docker and Kubernetes, and have shipped AI agent systems with function calling and RAG pipelines.

## TECHNICAL SKILLS

**Frontend:** React, Next.js, TypeScript, JavaScript, Vue.js, SvelteKit, React Flow, Three.js, Redux, Zustand, Tailwind CSS, SASS/SCSS, HTML5, CSS3, Responsive Design

**Testing:** Jest, React Testing Library, Playwright, E2E Testing, API Testing

**Backend:** Node.js, Express.js, FastAPI, Python, Django (familiar), Go (familiar), C#

**APIs & Authentication:** REST, GraphQL, JWT, RBAC, OAuth, SSO/OIDC (familiar), SAML (familiar)

**Databases:** PostgreSQL, MongoDB, MySQL

**DevOps & Infrastructure:** Git, Docker, Kubernetes (familiar), Helm, Terraform (familiar), GitHub Actions, AWS, GCP, CI/CD

**AI:** OpenAI API, Google GenAI, Function Calling, RAG, Multi-Agent Orchestration, MCP Servers, LLM Evals

## PROFESSIONAL EXPERIENCE

**Frontend & Full Stack Engineer** - Freelance *Feb 2026 – Present*

- Built and maintained reusable design systems and component libraries with React, TypeScript, and Tailwind CSS, cutting future development effort by 40% and improving frontend performance by 25% through optimised state management (Redux, Zustand) and lazy-loading.
- Embedded directly with 5+ clients to scope workflows and ship production software end-to-end React UI, REST API integration, deployment, and knowledge transfer on client infrastructure.
- Integrated enterprise authentication (JWT, RBAC, session management) and third-party APIs across multiple client frontends, improving development efficiency by 30% through reusable adapter patterns.
- Contributed to Aden Hive, an open-source AI agent platform, and instrumented client services with structured logging and error tracking, writing runbooks and component docs so teams could operate independently.

**Frontend Engineer** - BagYo (Startup) *Nov 2025 – Feb 2026*

- Built the BagYo logistics platform frontend in Vue.js and TypeScript, working directly with the founding team to translate product requirements into shipped features.
- Built and maintained 10+ reusable UI components across the dashboard and platform views, establishing a scalable frontend architecture that reduced duplication and sped up future feature work.
- Implemented end-to-end authentication (login, session management, role-based access) and integrated backend APIs to power core dashboard functionality.
- Worked cross-functionally with the BagYo team to debug and refine dashboard features based on direct product feedback, improving usability and reliability.

**Frontend Developer** - Linchpin Organization *Dec 2023 – Nov 2025*

- Led frontend development of the organisation's website in React, translating Figma mockups to production interfaces with 95% design accuracy and improving UX and mobile responsiveness by 35%.
- Built and maintained a reusable design system of 20+ React components, still in active production use, cutting future development effort by 40% and letting contributors onboard without assistance.
- Collaborated with cross-functional teams to deliver production-ready features on deadline documented component usage, design decisions, and integration guides for long-term maintainability.

**Web Development Intern** - Madadgaar Foundation *Jul 2023 – Sep 2023*

- Developed responsive web pages and converted Figma mockups into functional interfaces with 95% design accuracy, improving mobile compatibility by 30%.
- Assisted with debugging and frontend optimisation, reducing reported frontend issues by 20%.

## PROJECTS

**Ledger AI - Auditable AI Workspace** *Production link*

- Took Ledger AI from 0 to production scoped the product, designed the information architecture, and built the full workspace UI in Next.js, TypeScript, and Tailwind.
- Built an interactive replay timeline with Zustand for state management, letting users step through any run prompt, retrieved context, tool calls, output, and eval scores as an immutable, shareable trace.
- Built a multi-agent pipeline (Planner → Knowledge → Builder → Verifier) with structured outputs and function calling each agent exposes its rationale and confidence score in the UI.
- Built a RAG retrieval pipeline (pgvector embeddings, cross-encoder reranking) with an automated eval harness (RAGAS) scoring retrieval quality, faithfulness, and citation accuracy on every run.

**REPROVIA - Reproducible Computational Research Infrastructure** *Production link*

- Built a React and TypeScript frontend centred on a React Flow visual workflow builder kept in sync with a YAML workflow specification, edited through an embedded Monaco editor with schema validation and DAG processing.
- Architected a FastAPI/Python backend with an execution-adapter pattern (local/mock and Kubernetes adapters) that runs workflow steps as Kubernetes Jobs/Pods with namespace isolation, ConfigMaps/Secrets, resource limits, and health probes.
- Designed artifact storage with immutable, cryptographically fingerprinted provenance records (workflow/dataset version, git commit, container image/digest, parameters, execution metadata, checksums, timestamps), and implemented workflow reproduction that compares runs as MATCHED, DIFFERENT, or INCONCLUSIVE.
- Built RBAC auth and structured request-level logging, and deployed with Docker and Helm charts with CI/CD for build and release. Independently designed not affiliated with or modelled as a clone of any existing CERN system.

**ShieldIQ - Human Risk Intelligence** *Production link*

- Built ShieldIQ, a multi-tenant B2B SaaS frontend in SvelteKit and TypeScript, covering phishing simulation campaigns, employee security-awareness tracking, and executive security posture dashboards.
- Developed RBAC, campaign management workflows, and audit logging, plus a weighted human-risk scoring layer producing department-level risk heatmaps, backed by a Go, PostgreSQL, DynamoDB, and Terraform infrastructure.

**CloudHunt - Cloud Incident Investigation Studio** *Production link*

- Took CloudHunt from 0 to a deployed DFIR workbench for SOC analysts, built end-to-end in SvelteKit 5 with Runes-based state management ($state, $derived, $props) and TypeScript.
- Shipped a multi-cloud log normalisation engine (AWS CloudTrail/GuardDuty, Azure Activity Logs, GCP Audit Logs) with automated IOC extraction, an interactive MITRE ATT&CK Cloud Matrix heatmap, and a Cytoscape.js threat-graph visualiser for blast-radius analysis.

**AutoLife AI - Smart Daily Automation System** *GitHub link*

- Scoped, designed, and shipped AutoLife AI from 0 to production designed the data-rich dashboard UI in Figma and built it with React, FastAPI, and PostgreSQL, deployed on GCP.
- Built agent-callable task routing APIs with structured outputs and function calling and multi-agent orchestration with routing logic, reducing manual effort by 40% and improving automation accuracy by 30%.
- Deployed containerised services with Docker on GCP, with CI/CD via GitHub Actions, structured logging, and latency monitoring.

**Ubuntu Web Performance Lab** *GitHub link*

- Building a monitoring dashboard that tracks the health and performance of a web application and its Ubuntu server, structured as a monorepo (apps/, shared packages) styled with SASS/SCSS, with Docker and Nginx.
- Tested frontend workflows across expected and edge-case scenarios and set up a GitHub Actions CI pipeline for linting and type-checking across the codebase.

## CERTIFICATIONS

- Google Cloud Certificate - Google Cloud / AutoLife AI Project, 2026
- Full Stack MERN Certificate - Acmegrade, 2023

## EDUCATION

**Bachelor of Computer Applications** - Presidency University, Bengaluru
*Sep 2022 – Sep 2025*